

Instructions for completing meta-methods review codebooks

Two codebooks

For comprehensibility, we employ two codebooks, one to capture the procedural variables of a given IGT study and a second to capture the scoring.

Missing and ambiguous data is meaningful

It will not always be possible to understand how the IGT was exactly administered in a study reported in an article and how the data were actually scored. Capturing this uncertainty or omission is a key goal here.

One score one row

The codebook for scoring follows a one-score-one-row format. Articles can have multiple rows, and each row should be fully completed.

Coding IGT scores

As a score we define each separate numerical value that is generated from IGT data and can be utilised to evaluate IGT responses. We only count scores that are explicitly mentioned in a given article, not scores that are only implied to have been calculated. Also, we don't count components of scores as individual scores if they are not reported separately. Note that scores can be reported in the text as well as exclusively in plots or tables. We provide a list of hypothetical examples at the end of this document.

Intention agnostic

Scores are agnostic to usage intention. If the article states that it used two scores, two scores are recorded.

Separation of procedure and scoring methods

Both procedure and scoring can vary between articles. Here we are particularly interested in scoring. As such, our descriptions of the scoring are recorded in a way that separates the details of the procedure from the scoring method. For example, if a given study uses only the last 60 of 120 trials to calculate a score, the scoring method is recorded as a proportion (0.5) rather than a number of trials

(60). This way, the scoring method would be judged to be identical to another study that used the last 50 of 100 trials, even though the procedures differed.

General strategy

1. The general strategy for completing the coding is to read a given article and then fill out the two codebooks, starting with the codebook for the procedure. Notes on each column are listed in the second row, along with rules for the contingencies among the columns. Contingencies among columns occur exclusively in the scoring codebook.
2. In the codebook for the procedure, a row is filled in for each study and IGT version. In the codebook for scoring, we switch to filling in one row for each score. Data must be complete for all rows.
3. If you can't be sure how to code a particular variable because the information in a given article is missing, ambiguous or vague, code indeterminable.
4. Start by filling out the article info and procedural details in the codebook for the procedure.
5. In multi-study papers, only the last study is coded. In the case of more than one IGT variant within one study, all variants are coded in separate rows.
6. In the variable protocol you should determine whether the publication specifies to follow a certain protocol of the task, which has already been described elsewhere. Standard protocol here means that the publication clearly specifies to follow the protocol described in Bechara et al. (1994) or Bechara et al. (1999), respectively, the latter describing the original paradigm but as a computerised version.
7. Regarding the variables goal, instructions and feedback, you are asked to assess whether they are equivalent to the original task. For reference, see Bechara et al. (1999), where the instructions and feedback given to the participants are described in detail.
8. If there are outstanding questions or something that is unclear or requires discussion about the procedure, record this in the `notes_procedure` variable. If an article uses a procedure that generally violates the layout of the spreadsheet, record the details of the procedure in the `notes_procedure` variable.
9. Then turn to the codebook for scoring. Copy the contents of the `article_id`, `authors`, `year` and `variant` columns from the codebook for the procedure. Since multiple scores can occur in a given article, the contents of these columns are duplicated accordingly for each score.
10. Then, for the first score found for that article and study, complete the scoring columns. Start with `score_description`, a verbal description of the score. Note, that scores for each variant of the IGT should be coded separately.

11. Compare this description to previous ones to see if this score has been used before – it may need to be clarified or rewritten to be sure. This will be an iterative process.
12. Use the contents of this description to complete `scoring_approach`, which specifies one of a small number of general strategies. Most articles should conform to one of the examples given in the second row of the spreadsheet, but it is likely that there will be exceptions.
13. Use the `scoring_approach` variable to then determine which of `deck_selection_formula`, `money_formula`, or `choice_behavior_formula` should be elaborated, whose values are contingent on `scoring_approach`. I.e., no more than one of these three variables should be recorded as something other than “not applicable” or “indeterminable”.
14. Fill in the remaining scoring columns `n_trials` and `subset_trials`. The `n_trials` column is essentially a duplicate of the column with the same name in the codebook for the procedure. It is again needed here to calculate the used proportions of the task later in order to differentiate scores.
15. Additional descriptive information in order to briefly contextualise a given score can be recorded in the `note_scoring` column. This may involve interrelationships between scores, for example, when multiple scores are used for a learning index. Analyses of scores are not addressed.
16. If there are outstanding questions or something that is unclear or requires discussion about the scoring method, record this as well in the `notes_scoring` column. If an article uses a score that generally violates the layout of the spreadsheet, record the details in the `notes_scoring` column.
17. When all columns are complete, find the next score employed within that study and article. Duplicate the article information, increment the `strategy_count` column, and enter the new values for the scoring columns.
18. When no more scores are found for a given article, move on to the next article.
19. Periodically ensure that values are coherent between columns and adhere to the rules among them that are specified in the second column.
20. This is likely to be an iterative process. Rows may need to be revisited – especially the `score_description` and `score_id` variables – as you are exposed to more scores across papers. These descriptions may become more general or abstract, or you may realise that what you initially thought were distinct scores are in fact the same.

Examples of IGT studies and how to code them

Here we provide a list of hypothetical examples of IGT studies (which borrow from real articles) in order to illustrate how to record IGT procedures and scores.

Examples for coding IGT procedural variables

Example articles 1, 2, 3 and 4

Article 1 reports that the original paradigm was used and refers to Bechara et al. (1994) for details.

Procedure recorded: standard protocol; Bechara et al. (1994) is consulted to fill in the remaining columns on procedural details.

Article 2 reports that the original paradigm from Bechara et al. (1994) was used and no changes were made to the task apart from translating the task to Spanish.

Procedure recorded: standard protocol; Bechara et al. (1994) is consulted to fill in the remaining columns on procedural details; the translation to Spanish is noted in the notes_procedure column.

Article 3 reports using a version of the IGT similar to the procedure described by Bechara et al. (1994). Few procedural details are reported which don't deviate from the original paradigm.

Procedure recorded: indeterminable protocol; reported procedural details are recorded in the respective columns; remaining columns are set to indeterminable.

Article 4 reports using the IGT without explicitly stating which paradigm is used. Bechara et al. (1994) is quoted in parentheses after the first occurrence of the word IGT. No further information is given on the procedure of the IGT.

Procedure recorded: indeterminable protocol; remaining columns are set to indeterminable.

Note: If an article specifies that the original paradigm was used, the protocol column is set to standard protocol and the remaining columns are filled in accordingly. In some cases the wording might be too vague to be sure that no modifications were made. Also, a mere citation is often not enough to infer what protocol was actually followed. Hence, in these cases, only distinct information is recorded and missing information is coded as indeterminable.

Example articles 5 and 6

Article 5 reports that in the used IGT version advantageous decks pay \$50 and disadvantageous decks pay \$100. No further information on the version of the IGT is given.

Procedure recorded: reward_adv is set to 50 and reward_disadv is set to 100; the remaining columns are set to indeterminable.

Article 6 reports that in the used IGT version gains and losses are different for each card but advantageous decks usually pay around \$50 and disadvantageous decks usually pay around \$100. No further information on the version of the IGT is given.

Procedure recorded: reward_adv and reward_disadv are set to indeterminable; the remaining columns are set to indeterminable; the paragraph is quoted in the notes_procedure column.

Note: It is possible to set all procedural variables to indeterminable if the information is too ambiguous or vague. In article 6 there is neither enough information to determine the exact amounts of the rewards nor to identify any other relevant information on the IGT version.

Examples for coding IGT scores

Example article 1

The article reports results of analyses comparing the IGT net scores ($[C+D]-[A+B]$) within five blocks of trials. The results are also shown in a plot.

Scores recorded: 5 (block 1 $[C+D]-[A+B]$, block 2 $[C+D]-[A+B]$, block 3 $[C+D]-[A+B]$, block 4 $[C+D]-[A+B]$, block 5 $[C+D]-[A+B]$)

Note: Each reported score is recorded separately.

Example article 2

The article reports results of analyses comparing the IGT total net score ($[C+D]-[A+B]$) between two groups. It also reports counts of selections from individual decks (A, B, C, D) in a plot but does not report results of analyses of these.

Scores recorded: 5 (A; B; C; D; $[C+D]-[A+B]$)

Note: Scores can be reported exclusively in a plot. Scores are agnostic to usage intention; if 5 scores are reported, 5 scores are recorded.

Example article 3

The article reports results of analyses comparing the IGT net score $([C+D]-[A+B])$ between two groups. Does not report in text, plot, or tables anything to do with selections from individual decks.

Scores recorded: 1 $([C+D]-[A+B])$

Note: Components of scores (in this case deck choices from decks A, B, C and D) are not counted as scores, since they are not reported individually.

Example article 4

The article reports the probabilities of deck shifts after trials with a win and after trials with a loss in a table. Inferences are based on the probability of deck shifting after a loss.

Scores recorded: 2 (deck shifts after a win, deck shifts after a loss)

Note: Scores can be exclusively reported in a table. Scores are agnostic to usage intention; a score is recorded if reported, even if no inferences are based on it.

Example articles 5 and 6

Article 5 reports results of analyses comparing a learning effect between: difference in net scores $([C+D]-[A+B])$ between early trials of the task and late trials of the task between two groups.

Scores recorded: 1 (difference: late $[C+D]-[A+B]$ - early $[C+D]-[A+B]$)

Article 6 reports results of analyses comparing a learning effect between: difference in net scores $([C+D]-[A+B])$ between early trials of the task and late trials of the task between two groups. The analyses and inferences are based on the difference scores, but they include a plot of the early and late scores (separately).

Scores recorded: 3 (early $[C+D]-[A+B]$, late $[C+D]-[A+B]$, difference: late $[C+D]-[A+B]$ - early $[C+D]-[A+B]$)

Note: Individual scores can in some cases also be components of other scores. They are only counted as separate scores if reported individually.

Example articles 7 and 8

Article 7 reports results of analyses comparing IGT net scores in late trials (last 40% of trials, $[C+D]-[A+B]$) between two groups. It notes that this score was calculated from those 40% trials.

Scores recorded: 1 (last 40% $[C+D]-[A+B]$)

Article 8 reports results of analyses comparing IGT net scores in late trials (last 40% of trials, $[C+D]-[A+B]$) between two groups. It notes that this score was calculated as the average of the last two blocks of 20 trials but does not report in text, figures, or tables the components of this average score by block.

Scores recorded: 1 (last 40% $[C+D]-[A+B]$)

Note: These two examples are coded identically, since components of scores are not counted as scores if not reported individually.

References

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